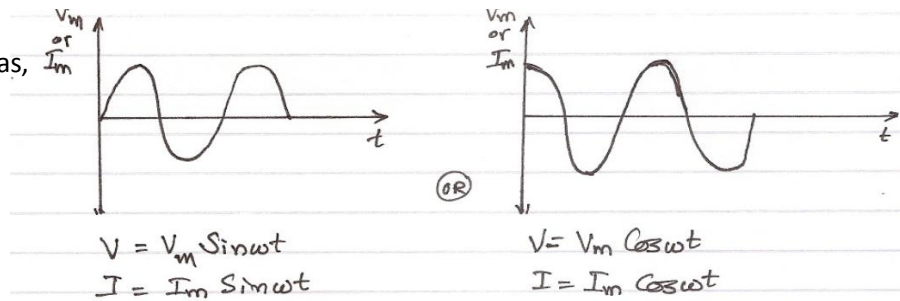


## NOTES ON 'ALTERNATING CURRENT'

Alternating current (A.C) is the current that changes in magnitude direction continuously with respect to time.

It can be represented as,



The currents and voltages in a.c circuits can be expressed by the following terms:-

- (i) **Instantaneous (I or V) :-** It is the current or voltage that in the circuit at any instant.
- (ii) **Peak ( $I_m$  or  $V_m$ ):-** It is the maximum available voltage or current in the circuit.
- (iii) **Average ( $I_{av}$  or  $V_{av}$ ):-** It is the arithmetic mean of the instantaneous values of voltages or currents in the circuit.
- (iv) **Root Mean Square (R.M.S) or Effective:-** It is the square root of the average of the squares of the instantaneous values of the voltages or currents in the circuit.

$$V_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots}{n}}$$

R.M.S value is the d.c equivalent of a.c

### Relation between Peak and R.M.S values

$$I_{rms}^2 = \int_0^T \frac{I^2 dt}{T} = \int_0^T \frac{I_m^2 \sin^2 \omega t dt}{T}$$

$$= \int_0^T \frac{I_m^2}{T} \left( \frac{1 - \cos 2\omega t}{2} \right) dt$$

$$\text{where } \omega = 2\pi/T$$

$$I_{rms}^2 = \frac{I_m^2}{2T} \left[ \int_0^T dt - \int_0^T \cos 2\omega t dt \right] \text{-----(1)}$$

$$\text{But } \int_0^T \cos 2\omega t dt = \left[ \frac{\sin 2\omega t}{2\omega} \right]_0^T = \frac{1}{2\omega} \left[ \sin \frac{4\pi}{T} t \right]_0^T = \frac{1}{2\omega} [\sin 4\pi - \sin 0] = 0$$

Thus equation (1) becomes,  $I_{rms}^2 = \frac{I_m^2}{2T} [T - 0]$  So,  $I_{rms} = \frac{I_m}{\sqrt{2}}$

Similarly, it can also be shown that,  $V_{rms} = \frac{V_m}{\sqrt{2}}$

**Note:** the measurement of ac is done by the comparison of the heating effect produced by ac with that by dc. Hence ac measuring instruments are also called 'hot-wire instruments'.

Show that the average current in the complete cycle of a.c is zero.

$$I_{av} = \int_0^T \frac{I dt}{T} = \int_0^T \frac{I_m \sin \omega t dt}{T} = \frac{I_m}{T} \int_0^T \sin \omega t dt = \frac{I_m}{T} \left[ \frac{-\cos \omega t}{\omega} \right]_0^T$$

$$= -\frac{I_m}{\omega T} \left[ \cos \frac{2\pi}{T} \right]_0^T = -\frac{I_m}{\omega T} [\cos 2\pi - \cos 0] = 0 \text{ (Hence proved)}$$

**Note:-** But in half a cycle,  $I_{av} = \int_0^{T/2} \frac{I dt}{T/2} = \int_0^{T/2} \frac{I_m \sin \omega t dt}{T/2} = 2 \frac{I_m}{T} \int_0^{T/2} \sin \omega t dt = 2 \frac{I_m}{T} \left[ \frac{-\cos \omega t}{\omega} \right]_0^{T/2}$

$$= -2 \frac{I_m}{\omega T} \left[ \cos \frac{2\pi}{T} \right]_0^{T/2} = -\frac{I_m}{\omega T} [\cos \pi - \cos 0] = -\frac{I_m}{\omega T} [-1 - 1]$$

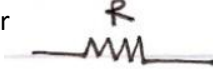
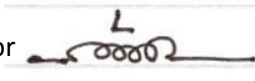
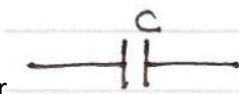
This gives  $I_{av} = \frac{2I_m}{\pi}$  Similarly, it can be shown that  $V_{av} = \frac{2V_m}{\pi}$

## Phasor diagrams

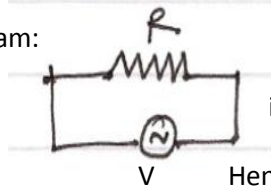
These are the vector diagrams in which the magnitudes represent the peak values and directions represent the phase differences between voltages and currents in the a.c circuits.

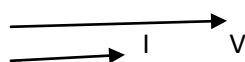
### A.C Circuits

Fundamental components in a.c circuits.

1. Resistor   $V = IR$
2. Inductor   $\mathcal{E} = -L \frac{dI}{dt}$
3. Capacitor   $Q = CV$

#### 1. Circuit containing only resistor

Circuit diagram:  Let  $V = V_m \sin \omega t$  Current ( $I$ ) =  $V/R = (V_m \sin \omega t)/R$   
i.e.,  $I = I_m \sin \omega t$ . Where  $I_m = V_m/R$ .  $R = V_m/I_m$   
Hence the phase difference between voltage and current is,  $\phi = 0$ .

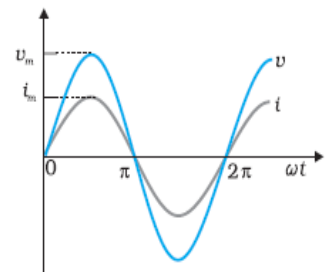
So the phasor diagram can be as shown 

The graph showing the variation of current and voltage with time can be represented as:

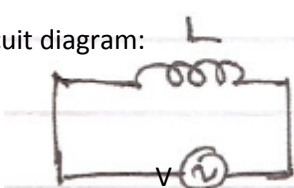
#### Power associated with the circuit:

$$P_R = \int_0^T \frac{VIdt}{T} = \frac{1}{T} \int_0^T V_m \sin \omega t \cdot I_m \sin \omega t dt = \frac{V_m I_m}{T} \int_0^T \sin^2 \omega t \cdot dt$$
$$= \frac{V_m I_m}{T} \int_0^T \left( \frac{1 - \cos 2\omega t}{2} \right) dt = \frac{V_m I_m}{T} \left[ \int_0^T \frac{1}{2} dt - \int_0^T \frac{\cos 2\omega t}{2} dt \right]$$

But  $\int_0^T \frac{\cos 2\omega t}{2} dt = 0$  Thus we get,  $P_R = (V_m I_m) / 2$  Hence  $P_R = V_{rms} \cdot I_{rms}$



#### 2. Circuit containing only inductor

Circuit diagram:  Let  $V = V_m \sin \omega t$   
The induced voltage in the inductor is  $V_{ind} = -L(dI/dt)$   
By loop rule,  $V_m \sin \omega t - L(dI/dt) = 0$   
i.e.,  $V_m \sin \omega t = L(dI/dt)$  this gives,  $dI = (V_m/L) \sin \omega t dt$

On integrating the above equation we get, Current ( $I$ ) =  $\frac{V_m}{L} \left( \frac{-\cos \omega t}{\omega} \right)$

$$I = -\frac{V_m}{\omega L} \cos \omega t \quad \text{Hence } I = I_m \sin (\omega t - \pi/2) \quad \text{where } I_m = (V_m/\omega L)$$

Here ' $\omega L$ ' represents the non-resistive opposition offered by the inductor to the flow of current in an ac circuit. It is called inductive reactance ( $X_L$ ) i.e.,  $X_L = \omega L$

Hence the phase difference between voltage and current is,  $\phi = \pi/2$ .

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Here the current lags behind the voltage by  $\pi/2$ .

So the phasor diagram can be as shown

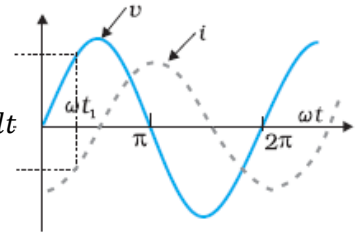


The graph showing the variation of current and voltage with time can be represented as:

#### Power associated with the circuit:

$$P_L = \int_0^T \frac{VI dt}{T} = \frac{1}{T} \int_0^T V_m \sin \omega t \cdot I_m (\sin \omega t - \pi/2) dt = \frac{V_m I_m}{T} \int_0^T \sin \omega t \cdot -\cos \omega t dt$$

$$= -\frac{V_m I_m}{T} \int_0^T \left( \frac{\sin 2\omega t}{2} \right) dt = -\frac{V_m I_m}{2T} \cdot 0 \quad \text{as} \quad \int_0^T \sin 2\omega t dt = 0$$

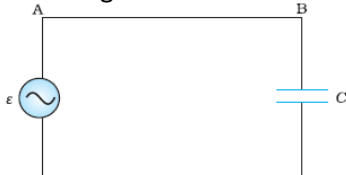


Thus we get,  $P_L = 0$  i.e., an ideal inductor does not consume any power.

(Note: this is why they are used as 'choke coils' in ac circuits to reduce current without power loss, in the place of resistors)

### 3. Circuit containing only capacitor

Circuit diagram:



Let  $V = V_m \sin \omega t$

We have  $Q = CV$

Differentiating we get,  $\frac{dQ}{dt} = C \frac{dV}{dt}$

i.e.  $I = C V_m \cos \omega t \cdot \omega \quad I = \frac{V_m}{1/C\omega} \sin(\omega t + \frac{\pi}{2})$

Hence

$$I = I_m \sin(\omega t + \pi/2)$$

$$\text{where } I_m = \frac{V_m}{1/C\omega}$$

Thus  $X_C = 1/C\omega$  gives the non-resistive opposition to the flow of current called capacitive reactance.

Hence the phase difference between voltage and current is,  $\phi = \pi/2$ .

Here the current **leads** the voltage by  $\pi/2$ . So the phasor diagram



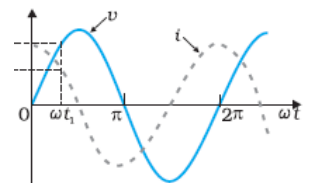
Is shown beside:-

The graph showing the variation of current and voltage with time can be represented as:

#### Power associated with the circuit:

$$P_L = \int_0^T \frac{VI dt}{T} = \frac{1}{T} \int_0^T V_m \sin \omega t \cdot I_m (\sin \omega t + \pi/2) dt = \frac{V_m I_m}{T} \int_0^T \sin \omega t \cdot \cos \omega t dt$$

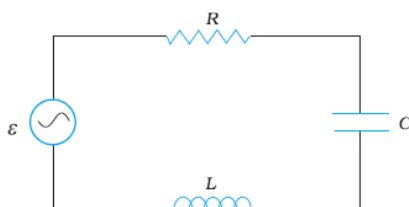
$$= \frac{V_m I_m}{T} \int_0^T \left( \frac{\sin 2\omega t}{2} \right) dt = \frac{V_m I_m}{2T} \cdot 0 \quad \text{as} \quad \int_0^T \sin 2\omega t dt = 0$$



Thus we get,  $P_C = 0$  i.e., an ideal capacitor does not consume any power.

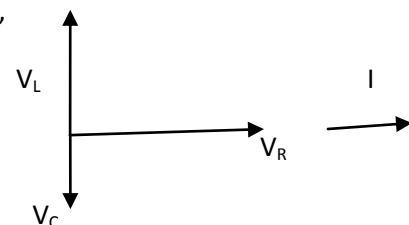
### 3. L-C-R series circuit

Circuit diagram:

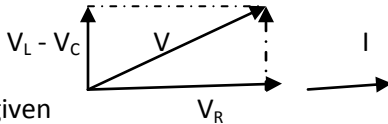


Let  $V = V_m \sin \omega t$

Since the components are in series, the current,  $I$  is the same throughout. Let  $V_R$ ,  $V_L$  and  $V_C$  be the p.d's across the resistor, inductor and capacitor respectively. Assuming  $V_L$  to be greater than  $V_C$ , the phasor diagram can be drawn as,



The resultant of the above vector diagram is:

By parallelogram law, 

The resultant voltage is given

by,  $V = \sqrt{V_R^2 + (V_L - V_C)^2}$  But  $V_R = IR$ ,  $V_L = IX_L$  and  $V_C = IX_C$

Thus  $V = I \sqrt{R^2 + (X_L - X_C)^2}$  Hence  $V/I = \sqrt{R^2 + (X_L - X_C)^2}$

This gives the total opposition to the flow of current in an a.c circuit, including resistance and reactances. It is called the **impedance (Z)** of the circuit.

Thus the impedance is given by  $Z = \sqrt{R^2 + (X_L - X_C)^2}$

Also the phase difference between the voltage and current is given by

$$\tan \phi = \frac{V_L - V_C}{V_R} \quad \text{i.e. } \tan \phi = \frac{(IX_L - IX_C)}{IR} \quad \text{or} \quad \tan \phi = \frac{(X_L - X_C)}{R}$$

Hence, current in the circuit can be expressed as  $I = I_m \sin(\omega t - \phi)$ , the negative sign indicates that the inductive reactance dominates capacitive. i.e. current lags behind voltage.

(Refer page 246 of Text book for 'Analytical solution')

#### Power associated with the circuit:

$$\begin{aligned} P &= \int_0^T \frac{VIdt}{T} = \frac{1}{T} \int_0^T V_m \sin \omega t \cdot I_m (\sin \omega t - \phi) dt = \frac{V_m I_m}{T} \int_0^T \sin \omega t [\sin \omega t \cdot \cos \phi - \cos \omega t \cdot \sin \phi] dt \\ &= \frac{V_m I_m}{T} \left[ \cos \phi \int_0^T \sin^2 \omega t \cdot dt - \sin \phi \int_0^T \sin \omega t \cdot \cos \omega t dt \right] \\ &= \frac{V_m I_m}{T} \left[ \cos \phi \int_0^T \left( \frac{1 - \cos 2\omega t}{2} \right) dt - \sin \phi \int_0^T \frac{\sin 2\omega t}{2} dt \right] \end{aligned}$$

This gives  $P = V_{rms} \cdot I_{rms} \cdot \cos \phi$ . (Refer above for the integration results)

It shows that power in the circuit depends not only the values of current and voltage, but on the value of 'cosφ' also. Hence it is called the power factor of the circuit.

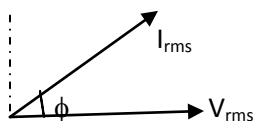
i.e. power factor,  $\cos \phi = (V_R/V) = (IR/IZ)$   $\cos \phi = \frac{R}{Z}$

Thus power can also be expressed as  $P = V_{rms} \cdot I_{rms} \cdot \frac{R}{Z}$

**Wattless current:-** It is the current in the a.c circuit in which there is a phase difference of 'π/2' between current and voltage.

We have  $P = V_{rms} \cdot I_{rms} \cdot \cos \phi$  Here  $\cos \phi = 0$

It can be represented as



Thus by resolving the current into the respective components, we get the

Wattless current =  $I_{rms} \cdot \sin \phi$

**RESONANCE:-** It is the condition at which the inductive and capacitive reactances are equal so that the current in the circuit is maximum.

We know  $I_m = V/Z$  Current is maximum, when impedance is minimum.

But  $Z = \sqrt{R^2 + (X_L - X_C)^2}$  So 'Z' is minimum when  $X_L = X_C$

Hence  $Z_{min} = R$

$\omega L = 1/\omega C$  or  $\omega^2 = 1/LC$

Thus  $(I_m)_{\max} = V_m/R$

Resonant frequency  $\omega_r = \frac{1}{\sqrt{LC}}$  and  $\nu_r = \frac{1}{2\pi\sqrt{LC}}$

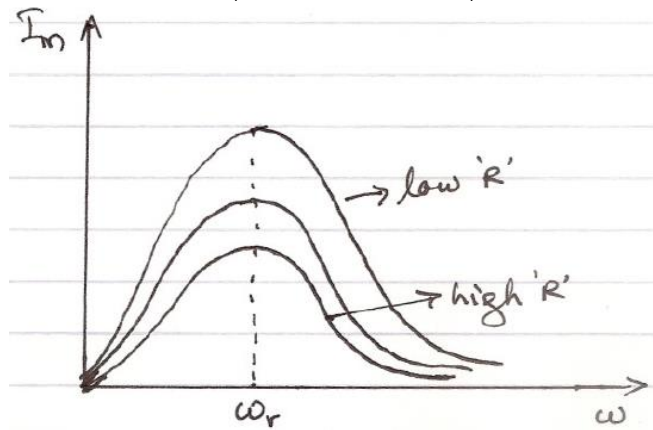
Frequency response curves with current

The variation of peak current

with different frequencies

can be represented for different values

of resistances as shown beside:

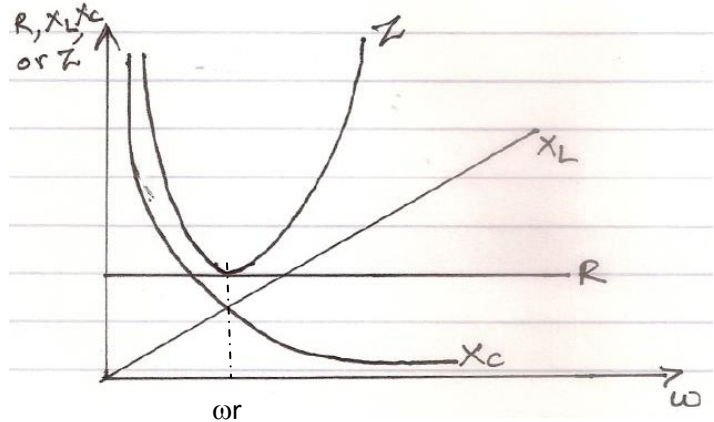


### VARIATION OF RESISTANCE, REACTANCE AND IMPEDANCE WITH FREQUENCY

Resistance is independent of the frequency.

$$X_C = 1/C\omega \quad \text{So, } X_C \propto 1/\omega$$

$$X_L = \omega L \quad X_L \propto \omega$$



### POWER AT RESONANCE

We have Power  $P = V_{\text{rms}} \cdot I_{\text{rms}} \cdot \frac{R}{Z}$  But  $V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$ ,  $I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$  and  $I_m = V_m/Z$

$$\text{So } P = \frac{V_m^2}{2Z^2} R \quad \text{----- (1)}$$

At resonance, Z is minimum. Thus power also is maximum at resonance.

$$\text{Hence } P_{\max} = \frac{V_m^2}{2R} \quad \text{Eqn(1) becomes } P = P_{\max} \frac{R^2}{Z^2}$$

The frequency response curve with power can be drawn as:

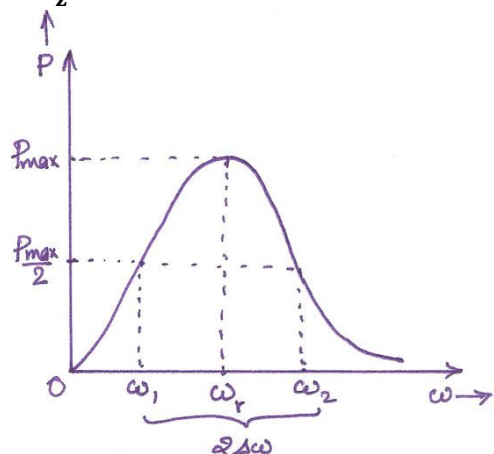
The frequencies at which the power in the circuit is half of

Its maximum value are called '**half power points**'.

The difference between half power points is called

### **Bandwidth ( $2\Delta\omega$ )**

$$2\Delta\omega = \omega_2 - \omega_1$$



### Q-FACTOR (Quality factor)

It measures the sharpness of the resonance curve. It is defined as the ratio of the resonant frequency to the band width.

$$\text{i.e. } Q = \frac{\omega_r}{2\Delta\omega} \quad \text{Using the above equations we get,}$$

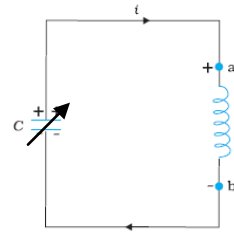
$$Q = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} \quad \text{so, } Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

## Radio Tuner

Tuning is the process of receiving the frequency of a desired radio station. It consists of an inductor and a variable capacitor.

During tuning, the value of capacitance is changed. At a given position the value of capacitance is such that it satisfies for a given frequency in the

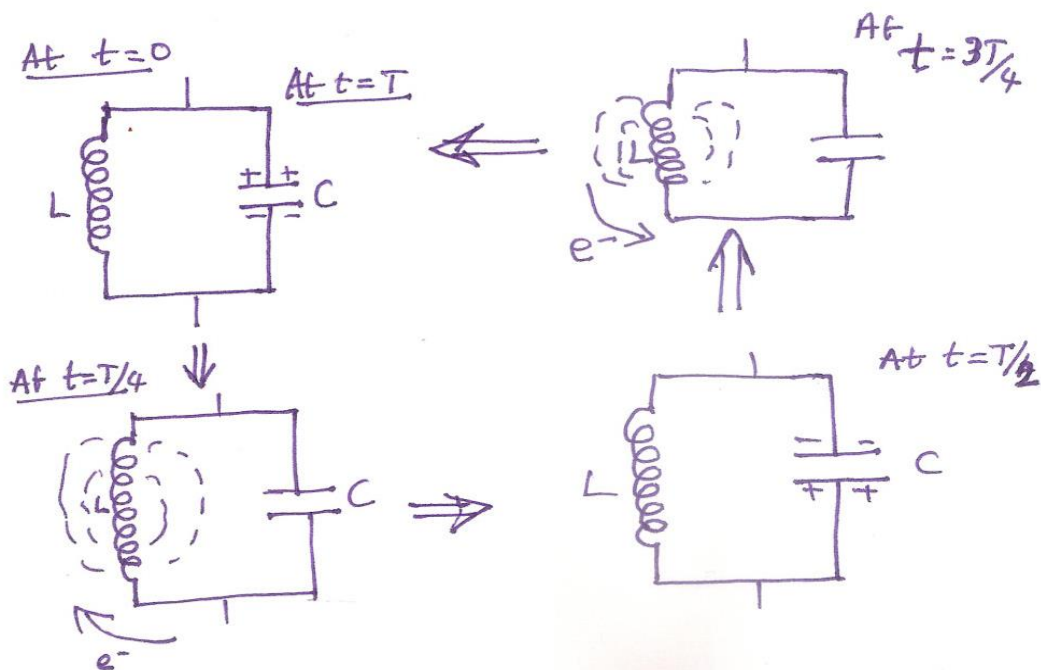
expression for resonant frequency,  $\nu_r = \frac{1}{2\pi\sqrt{LC}}$



Since the Q-factor of the circuit is adjusted to be very high, the maximum power reception occurs only at this frequency. Thus tuning is achieved.

## L-C OSCILLATIONS

across



connected

across:

## Mathematical Treatment

[Qn. Show that electric charges execute S.H.M in an L-C circuit. Hence derive an expression for the frequency of oscillations. Show that the total energy in the circuit is conserved.]

## Advantages of A.C

- (i) Cost of production is low.
- (ii) It can be transmitted to distant places with minimum power loss, with the help of transformers.
- (iii) It can easily be converted into d.c, wherever needed.
- (iv) Current in the circuits can be controlled with the help of choke coils.

## Disadvantages of A.C

- (i) It is more dangerous.
- (ii) It cannot be used in the processes like electroplating, making permanent magnets, etc.
- (iii) Energy loss can occur in the case of high frequency a.c due to **skin effect**. [ It is the effect in which high frequency a.c confines through the surface of a conductor, when transmitted.]